

Supplementary Material

Learning Utilities For High Utility Itemsets Mining

1 Complementary results for the aggregated regret values

Figures 1 plots some complementary comparison of TKO, TKU-CE and HAISAMPLER aggregated regrets for different query sizes w.r.t. *maximal* quality, with $\ell = 1$.

2 Complementary results for cumulative and non-cumulative

Figures 2 and 3 plot a detailed view of comparison between TKO, TKU-CE and HAISAMPLER (cumulative and non-cumulative regret) for different datasets w.r.t. *maximal* quality, with $k = 7$ and $\eta = 0.13$.

3 Complementary results for $Regret_{Avg}$ and $Regret_{Max}$ values

Tables 1 to 2 give detailed values of the regret for each dataset. Each table reports results for each combination of features : I $\sim$ Items, IT $\sim$ Items and Transactions.

4 Complementary results for CPU-times comparison

Figure 4 compares the CPU-times of TKO, TKU-CE and HAISAMPLER w.r.t. the I feature combination, with $k = 7$, $\ell = 0$ and $\eta = 0.13$.

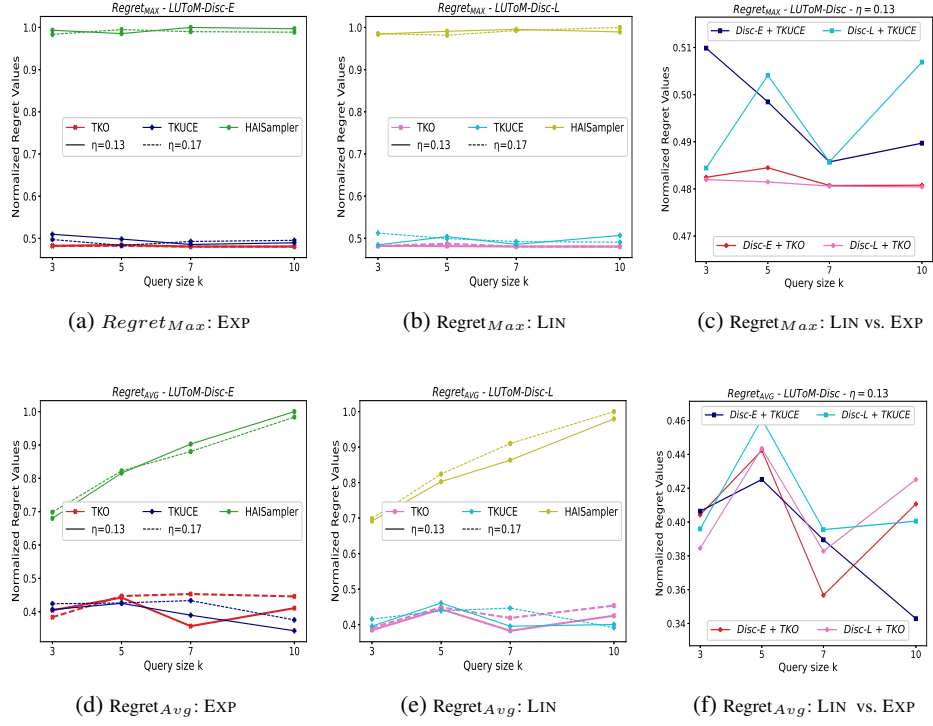
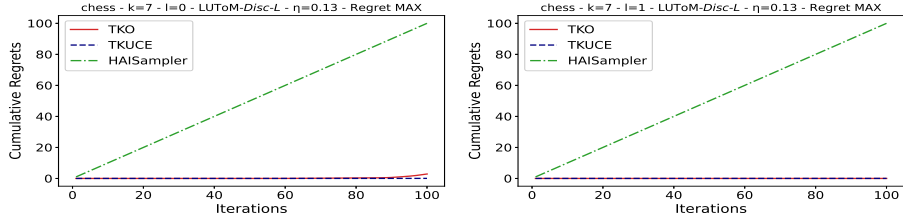
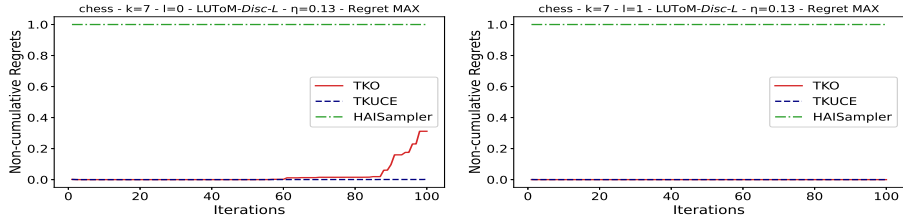
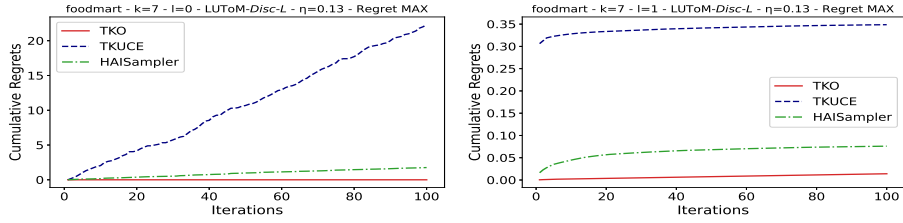
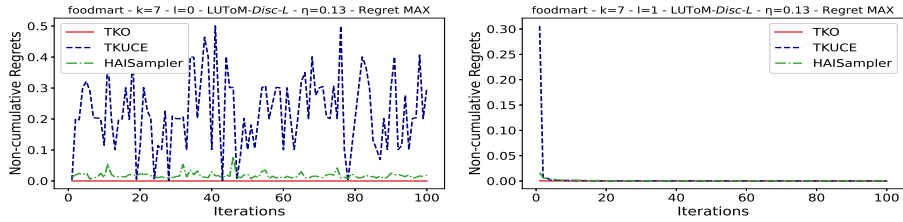
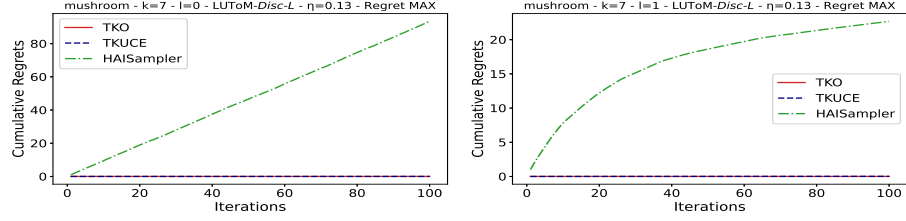
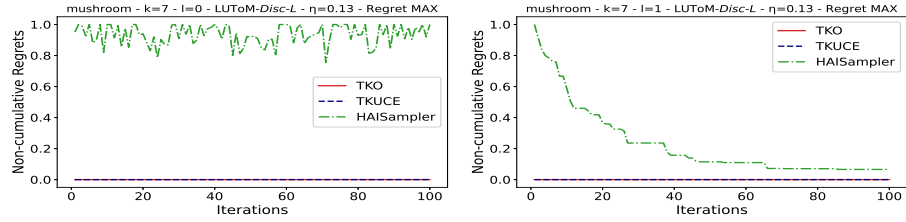


Fig. 1: Effects of the parameters η and k on $Regret_{Avg}$ and $Regret_{Max}$ with $\ell = 1$. Results are aggregated over all datasets and the two features combination (I, IT). Regret values are normalized to the range $[0, 1]$ based on the maximum regret value. Lower values are better.

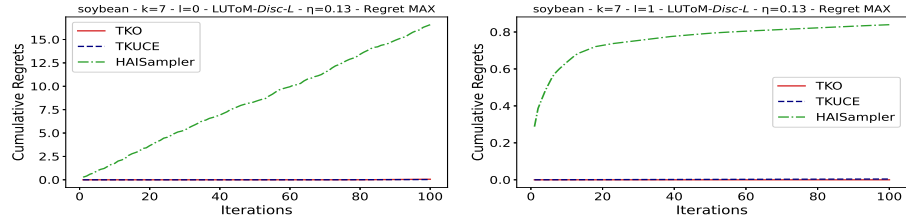
(a) CHESS: cumulative $Regret_{Max}$ ($\ell \in \{0, 1\}$).(b) CHESS: non cumulative $Regret_{Max}$ ($\ell \in \{0, 1\}$).(c) FOODMART: Cumulative $Regret_{Max}$ ($\ell \in \{0, 1\}$).(d) FOODMART: non cumulative $Regret_{Max}$ ($\ell \in \{0, 1\}$).Fig. 2: Detailed view of LUTOM-DISC-L results for $k = 7$ and $\eta = 0.13$.



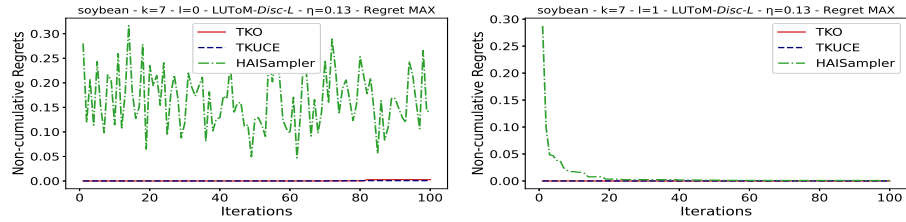
(a) MUSHROOM: Cumulative $Regret_{Max}$ ($\ell \in \{0, 1\}$).



(b) MUSHROOM: non cumulative $Regret_{Max}$ ($\ell \in \{0, 1\}$).



(c) SOYBEAN: cumulative $Regret_{Max}$ ($\ell \in \{0, 1\}$).



(d) SOYBEAN: non cumulative $Regret_{Max}$ ($\ell \in \{0, 1\}$).

Fig. 3: Detailed view of LUTOM-DISC-L results for $k = 7$ and $\eta = 0.13$.

Table 1: Evaluation of features I and comparison of LUToM-DISC-L ($\eta = 0.13$) with LUToM-WO-DISC for different oracles. Results are given for each dataset ($k = 7$): (1) TKO, (2) TKU-CE and (3) HAISAMPLER.

(a) Regret values obtained by LUToM-DISC-L

Datasets	$\ell = 0$						$\ell = 1$					
	$Regret_{Max}$			$Regret_{Avg}$			$Regret_{Max}$			$Regret_{Avg}$		
	(1)	(2)	(3)	(1)	(2)	(3)	(1)	(2)	(3)	(1)	(2)	(3)
chess	0.312	0.002	1.000	0.408	0.121	1.000	0.000	0.000	1.000	0.415	0.501	1.000
foodmart	0.000	0.301	0.019	0.088	0.423	0.420	0.000	0.000	0.000	0.247	0.277	0.137
german-credit	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
mushroom	0.000	0.000	1.000	0.047	0.120	1.000	0.000	0.000	0.065	0.066	0.095	0.836
soybean	0.003	0.001	0.142	0.069	0.005	0.584	0.000	0.000	0.001	0.045	0.002	0.519
vote	0.070	0.014	0.178	0.156	0.082	0.789	0.001	0.014	0.002	0.310	0.115	0.572

(b) Regret values obtained by LUToM-WO-DISC

Datasets	$\ell = 0$						$\ell = 1$					
	$Regret_{Max}$			$Regret_{Avg}$			$Regret_{Max}$			$Regret_{Avg}$		
	(1)	(2)	(3)	(1)	(2)	(3)	(1)	(2)	(3)	(1)	(2)	(3)
chess	0.000	0.000	1.000	0.003	0.002	1.000	0.000	0.000	1.000	0.001	0.005	1.000
foodmart	0.000	0.201	0.016	0.001	0.224	0.330	0.000	0.000	0.000	0.009	0.267	0.232
german-credit	0.001	0.099	1.000	0.003	0.264	1.000	0.000	0.008	1.000	0.003	0.033	1.000
mushroom	0.000	0.000	0.825	0.001	0.004	0.975	0.000	0.000	0.072	0.002	0.004	0.837
soybean	0.000	0.000	0.155	0.001	0.002	0.652	0.000	0.000	0.002	0.002	0.004	0.365
vote	0.000	0.004	0.255	0.004	0.011	0.730	0.001	0.014	0.002	0.003	0.039	0.633

Table 2: Evaluation of features IT and comparison of LUTOM-DISC-L ($\eta = 0.13$) with LUTOM-WO-DISC for different oracles. Results are given for each dataset ($k = 7$): (1) TKO, (2) TKU-CE and (3) HAISAMPLER.

(a) Regret values obtained by LUTOM-DISC-L

Datasets	$\ell = 0$						$\ell = 1$					
	$Regret_{Max}$			$Regret_{Avg}$			$Regret_{Max}$			$Regret_{Avg}$		
	(1)	(2)	(3)	(1)	(2)	(3)	(1)	(2)	(3)	(1)	(2)	(3)
chess	0.000	0.001	1.000	0.010	0.011	1.000	0.000	0.000	1.000	0.011	0.005	1.000
foodmart	0.409	0.559	0.013	0.754	0.802	0.238	0.000	0.001	0.000	0.554	0.720	0.144
german-credit	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
mushroom	0.001	0.001	0.918	0.003	0.003	0.988	0.000	0.000	0.072	0.010	0.008	0.853
soybean	0.000	0.000	0.219	0.002	0.002	0.716	0.000	0.000	0.002	0.001	0.002	0.612
vote	0.001	0.044	0.348	0.007	0.114	0.747	0.000	0.008	0.003	0.015	0.071	0.613

(b) Regret values obtained by LUTOM-WO-DISC

Datasets	$\ell = 0$						$\ell = 1$					
	$Regret_{Max}$			$Regret_{Avg}$			$Regret_{Max}$			$Regret_{Avg}$		
	(1)	(2)	(3)	(1)	(2)	(3)	(1)	(2)	(3)	(1)	(2)	(3)
chess	0.000	0.000	1.000	0.002	0.002	1.000	0.000	0.000	1.000	0.002	0.002	1.000
foodmart	0.000	0.501	0.022	0.106	0.507	0.239	0.000	0.000	0.000	0.022	0.256	0.220
german-credit	0.000	0.046	1.000	0.005	0.124	1.000	0.000	0.016	1.000	0.006	0.085	1.000
mushroom	0.000	0.000	0.970	0.001	0.001	0.996	0.000	0.000	0.052	0.003	0.001	0.834
soybean	0.000	0.000	0.142	0.001	0.001	0.658	0.000	0.000	0.001	0.000	0.001	0.587
vote	0.000	0.018	0.127	0.001	0.041	0.606	0.000	0.008	0.002	0.002	0.020	0.636

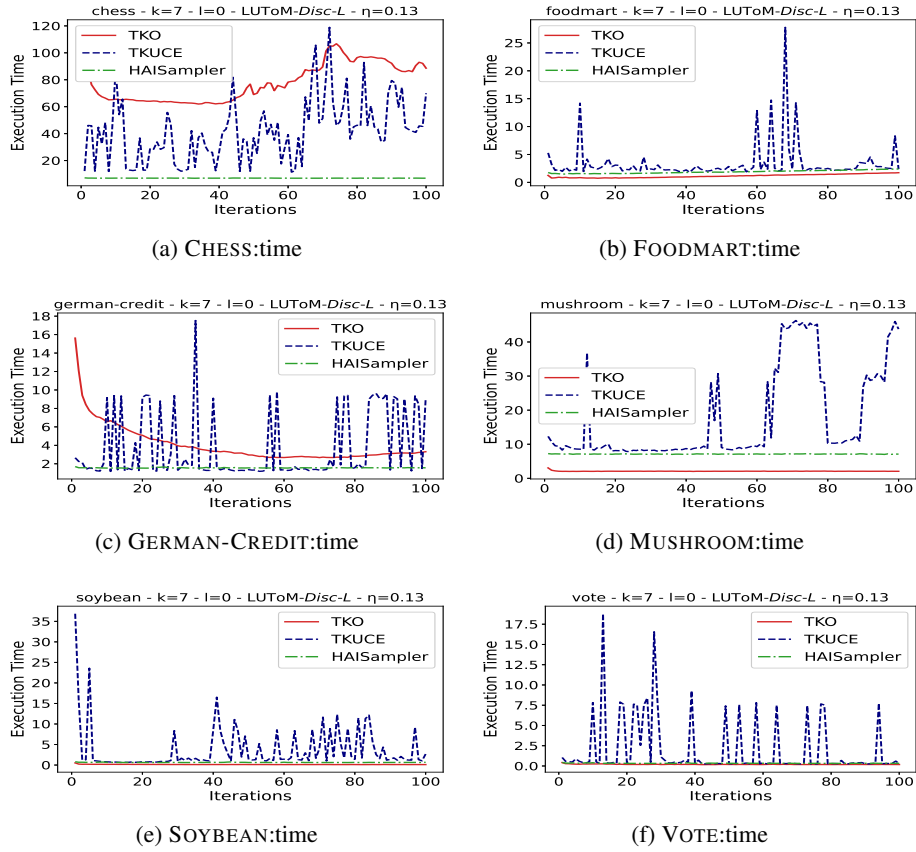


Fig. 4: Comparing the CPU-times of TKO, TKU-CE and HAISAMPLER w.r.t. the I feature combination with $k = 7$ and $\eta = 0.13$.